

Session Objectives

- 1. Students will be able to understand why the name “Short Leg Syndrome” is used to date, what symptoms and/or findings comprise the syndrome, and understand true and apparent leg-length discrepancy. Students will be able to understand what anatomical landmarks are measured and considered with radiological postural evaluation
- 2. Students will be able to understand a "C-scoliotic response" and an "S-scoliotic response." Students will be able to understand and review the bony, ligamentous and muscular involvement in the lumbo-sacral-pelvic area, in relation to the “Short Leg Syndrome.”
- 3. Students will be able to understand how to organize the evaluation and examination for the "Short Leg Syndrome," including Special Tests. Students will be able to understand what heel lift therapy is, and to which side a heel lift would be added.

Session Objectives

- 4. Students will be able to understand how to utilize David Heilig's formula, for calculation of heel lift height based on the radiologically measured sacral base unleveling (SBU). Students will be able to consider differential diagnoses potentially contributing to the symptoms of “Short Leg Syndrome.”
- 5. Students will be able to understand the maximum height of heel lift that can be placed inside of a shoe, and how the heel lift height added to the outside of the shoe is managed. Students will be able to understand the goals of treatment for “Short Leg Syndrome.”

“Short Leg Syndrome”

- A misnomer, really. Name used due to historical precedent.
- Actual causes may be unrelated to leg lengths.
- A “syndrome” since there are multiple symptoms and findings.
 - Sacral base unleveling
 - Innominate rotation (compensatory)
 - Caudal curve side-bending away from side of low sacral base, most often.

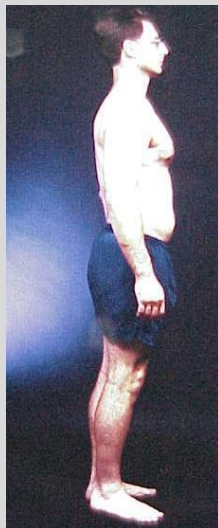
“Short Leg Syndrome”

- Diagnosis and Treatment

- Treat all somatic dysfunction diagnoses as well as possible **before** sending patient for radiological evaluation.
- Radiology - Measure coronal values of iliac crest heights, femoral head heights, sacral base unleveling, compensatory scoliotic curves (type, location, degree).
- Remember: 2mm human error in measurement; 25% bony distortion.

Observation of Stance

- Iliac crest heights level
- ASIS & PSIS level with contralateral side
- Increased lordosis of the lumbar spine
- Orientation of knees and feet



Adams Forward Bend Test



- Pt. stands in front of physician with feet together.
- Ask pt. to round her/his head forward & bend forward.
- Physician is looking to see if there is a difference between the right and left side in the area of the rib cage (looking for a rib hump).
- Sample findings: in a pt. who is SB left & rotated right → right rib cage would be more posterior compared to the left side.
- Then, ask patient to side bend towards the side of the rib hump (convexity). If the rib hump disappears, the curve is functional, if it remains, the curve is more structural in nature.

“Short Leg Syndrome”

- David Heilig’s Formula:

$$L \leq [SBU] / [D+C]$$

- Duration (D)

- 1 = 0-10 years
- 2 = 10-30 years [→ 11-29 years]
- 3 = 30+ years

- Compensation (C)

- 0 = none observed
- 1 = sidebending and rotation
- 2 = wedging, facet size changes, endplates with horizontal growths, spurring

- Sacral Base Unleveling (SBU)

- Lift Required (L)

“Short Leg Syndrome”

Case 1: A patient has healed from a right femoral fracture, right femur is now longer than the left femur, and he has 1/2 inch sacral base unleveling (SBU). The new asymmetry is of minimal duration, and the patient has no compensatory changes. Using the Heilig Formula, how much heel lift would you provide, and on which side?

- A 1/2 inch, right side
- B 1/2 inch, left side
- C 3/4 inch, right side
- D 3/4 inch, left side
- E 7/8 inch, right side
- F 7/8 inch, left side

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“Short Leg Syndrome”

Case 2: A patient is 35 years old, and had a right lower extremity injury in early youth at age 10 which led to his 1/2 inch right lower sacral base unleveling (SBU). He has minimal compensation with side bending and rotation changes noted. Using the Heilig Formula, how much of a heel lift, and on which side, would you prescribe for this patient?

- A 1/8 inch, right side
- B 1/8 inch, left side
- C 1/6 inch, right side
- D 1/6 inch, left side
- E 1/4 inch, right side
- F 1/4 inch, left side

Sample Heilig Formula Calculations

- Case 2: Patient age 35, injured in early youth at age 10, minimal compensation with side bending and rotation changes noted
- $L = [SBU]/[D+C] = \frac{1}{2}''/[2 + 1] = \frac{1}{2}''/3 = 1/6''$

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Case 3: A 75-year-old patient had an injury in his youth at age 15 that led to his current 1/2 inch left lower sacral base unleveling (SBU). He has lumbar vertebral endplates with horizontal growths and spurring, marked rotation. How much of a heel lift would you prescribe, and on which side?

- A 7/8 inch, left side
- B 7/8 inch, right side
- C 1/10 inch, left side
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“Short Leg Syndrome”

Case 4: A 26-year-old patient had polio, which affected her right leg, in her youth at age 12. She has 1/2 inch right lower sacral base unleveling, and no compensatory changes. How much of a heel lift would you prescribe, and on which side?

- A 1/2 inch, left side
- B 1/2 inch, right side
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“Short Leg Syndrome”

- The standing flexion test should become negative, once a proper heel lift has been put in place, and the patient has no more pelvic or lower extremity dysfunctions.
- Maximum lift inside a shoe is $\frac{1}{4}$ inch.
- Maximum of $\frac{1}{2}$ inch between patient's heel and the floor. Variable placement of lift materials is possible.

“Short Leg Syndrome”

- Compressive forces can cause a bone to grow faster.
- Therefore, in a growing child, a lift under a short leg may cause that leg to grow faster.
- Important for the physician to monitor the heel lift treatment in children carefully.
- Anterior Lift therapy: affects pelvic rotation – to the same side as the lift.
- Overall Goal – reeducate patient towards an ideal posture.

Treatment of “Short Leg Syndrome”

- OMT – relax & stretch muscles, correct somatic dysfunction, increase mobility
- Exercise – to stretch & tone
- Lift therapy – based on sacral base unleveling, length of time present, amount of compensation

Summary Slide

- Introduction to “Short Leg Syndrome” and heel lift therapy, with consideration of the potentially involved anatomical areas.
- Practice calculating heel lift height with David Heilig’s formula.

Questions for Review

2. "Short Leg Syndrome" is a syndrome, since it has multiple symptoms and findings, such as

- A Sacral base unleveling, cervical spine anterior translation, bilateral tibial torsion
- B Sacral base leveling, innominate flaring, cranial curve side-bending away from the innominate outflaring
- C Sacral base unleveling, compensatory innominate rotation, caudal curve side-bending away from the side of the low sacral base in most cases
- D Sacral base leveling, bilateral femoral anteversion compensation in all cases, medial longitudinal arches decreased in all cases

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Questions for Review

3. What possible vertebral responses to a postural asymmetry, whether from an actual short leg or from a postural imbalance/somatic dysfunction, can occur?

Questions for Review

10. The lumbosacral compensation in "Short Leg Syndrome" or postural asymmetry always involves caudal-curve side bending away from the side of the low sacral base

- A True
- B False

Questions for Review

10. The lumbosacral compensation in "Short Leg Syndrome" or postural asymmetry always involves caudal-curve side bending away from the side of the low sacral base

- A True
- ***B False**